

GLOBAL ECONOMICS ANALYST

The Path to 2075 — Is Convergence Really Over?

- Do poor economies grow faster than rich economies? Many economists argue that the evidence for convergence between low- and high-income economies is weak, suggesting that generalized convergence was largely confined to the ‘hyper-globalization’ decade of 2000–10. If true, this would be bad news for global well-being, and it would fundamentally weaken a key justification for long-term investment in emerging markets.
- We argue that previous studies that reported only limited evidence of convergence are incomplete, because they fail to weight countries by population size – effectively treating China and Cyprus’s experiences as equally significant. With this simple adjustment, we find that economic convergence has been a consistent feature of the global economy since the 1980s.
- The key drivers of this convergence have been Asia’s most populous nations: China (since the early 1980s), and India, Indonesia and Bangladesh (from the late 1990s onwards). Together, these four countries account for more than 40% of the global population.
- Among developed economies, the US has been an important positive outlier. Despite possessing one of the world’s highest levels of GDP per capita, its GDP per capita growth has continued to outpace most other developed markets.
- The contrast that we find with unweighted convergence studies suggests an intuitive point: periods of rapid globalization, such as 2000–10, are especially beneficial for the development of small economies with limited domestic markets. However, for large emerging market economies – especially those in East Asia – economic convergence appears relatively robust. This ongoing convergence implies that the share of global GDP represented by emerging markets (and East Asian EMs in particular) will continue to rise over time.
- Looking ahead, risks exist in both directions. The development of AI seems likely to bolster the strong performance of the US and parts of East Asia (in particular, China, South Korea and Taiwan). On the downside, increased protectionism could result in a reversal of globalization, while climate change poses a particular risk to Southern Asia and parts of Africa.

Kevin Daly
+44(20)7774-5908 | kevin.daly@gs.com
Goldman Sachs International

The Path to 2075 — Is Convergence Really Over?

Economic Convergence, Led by Asia's Powerhouses

In this *Global Economics Analyst*, we review the performance of GDP per capita across 121 developed (DM) and emerging market (EM) economies since 1980, with a particular emphasis on the process of long-term economic convergence.

The principle of convergence – that poorer economies generally grow faster than richer economies over time – is a central assumption underpinning our *Path to 2075 projections* for the global economy (see Box 1).¹ The logic is that the marginal return on investment should be higher in poorer economies where capital is relatively scarce, which attracts investment, increases capital stock, and raises living standards.

We begin by outlining some key stylized facts about GDP per capita levels and growth rates across the world. *Exhibit 1* (left-hand side) depicts GDP per capita across various regions since 1980, while the right-hand panel presents the same data indexed to 1980. Several features stand out:

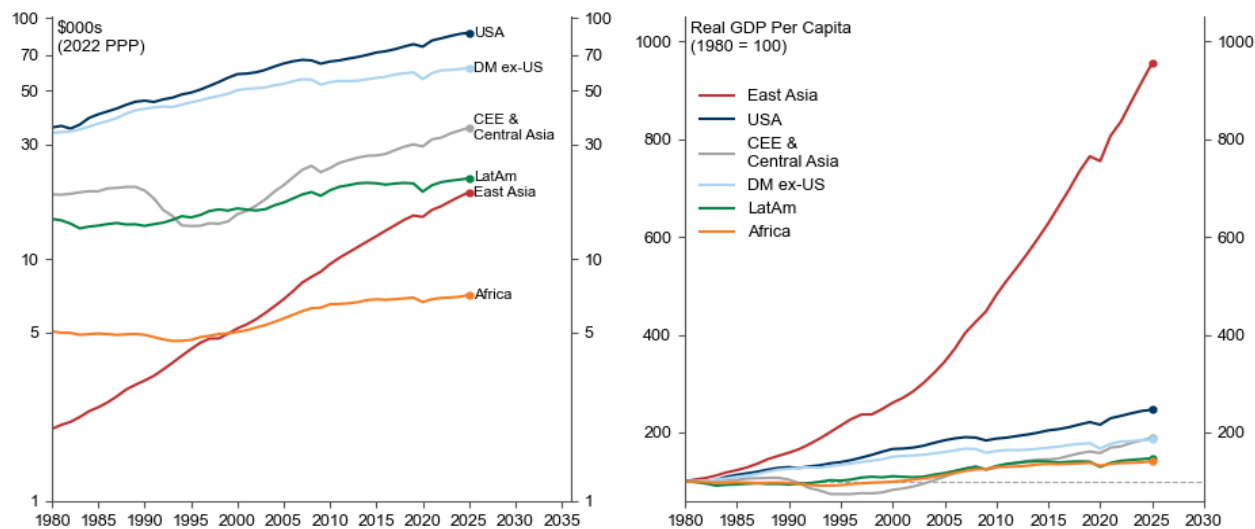
1. Cross-country differences in GDP per capita levels are large, even after adjusting for differences in purchasing power parity.² To take one striking example, GDP per person in the US is more than 12 times higher than the average across Africa.
2. East Asia's emerging markets have seen rapid GDP per capita growth, with average output per person now nearly 10 times higher than in 1980. However, GDP per capita in the region remains well below US and Western European levels, suggesting further potential for convergence.
3. US GDP per capita growth since 1980 has either matched or exceeded GDP per capita growth in the majority of other regions (with East Asia the only exception), despite having a much higher starting level of GDP per capita.
4. GDP per capita growth in Latin American and African economies has disappointed, averaging +0.9% and +0.8% per year respectively since 1980.
5. The break-up of the Soviet Union and the dissolution of the Eastern Bloc was initially very negative for GDP per capita in Eastern Europe and Central Asia in the early 1990s. However, GDP per capita levels have since rebounded in this region.

¹ “*The Path to 2075 — Slower Global Growth, But Convergence Remains Intact*”, Goldman Sachs Global Economics Paper, 6 December 2022.

² The choice between PPP weights and market weights depends on the comparison's purpose. PPP weights account for lower prices in poorer economies and better reflect welfare gains, while market weights show the Dollar value of each economy's GDP without adjusting for spending power. Since we are focused on income convergence, we use PPP weights. See “*How Fast Does the World Grow?*”, *GS Global Economics Analyst*, 27 October 2021.

Exhibit 1: GDP per Capita in East Asian EMs Has Risen Nearly Tenfold Since 1980, But Remains Well Below DM Levels

LHS: Real GDP Per Capita (\$000s log scale; 2022 PPP); RHS: Real GDP Per Capita (2022 PPP, 1980=100)



Regions are weighted by population size

Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

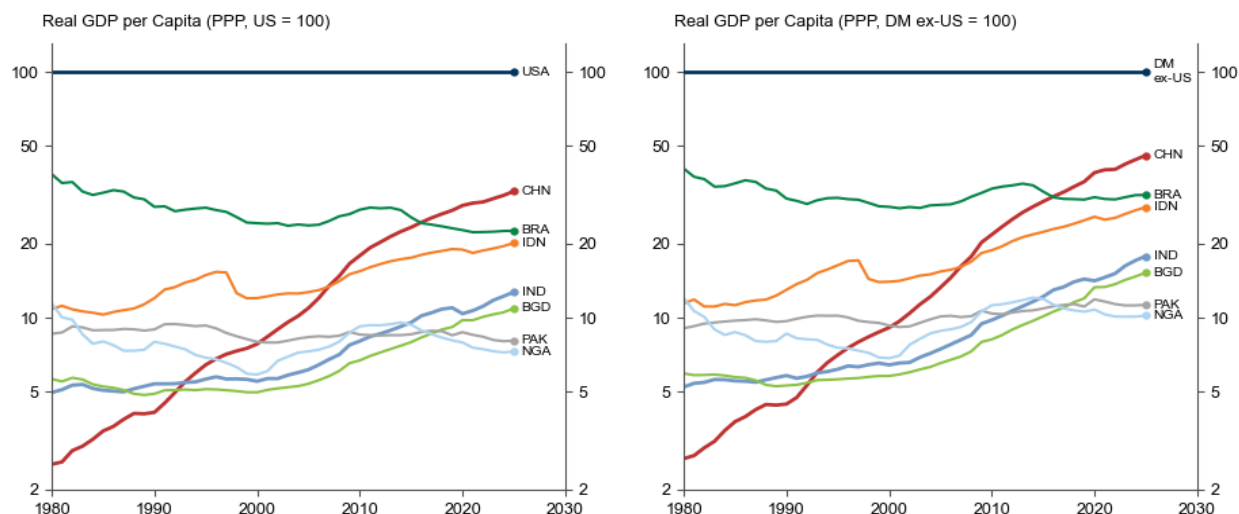
Exhibit 2 compares the GDP per capita of the eight most populous countries. The US is fixed as the ‘frontier economy’ in the left panel, while developed economies (excluding the US) are set as the benchmark in the right panel. China has seen significant income convergence, rising from just over 2% of US levels in 1980 to around 33% today. Since the late 1990s, India, Indonesia, and Bangladesh have also shown clear progress.

When compared to US GDP per capita, Brazil, Pakistan, and Nigeria have underperformed, with Brazil slipping further behind the US over time, while Pakistan and Nigeria have remained largely unchanged. However, as the US has shown particularly strong performance among developed markets, the relative picture shifts if we compare these economies to developed markets more broadly – Pakistan and Nigeria now show some convergence, whereas Brazil has remained more or less flat.³

³ The aim of this piece is to record, rather than explain, the convergence that has occurred. However, the question remains as to why the GDP per capita performance of Brazil, Pakistan and Nigeria has been disappointing. In Brazil's case, macroeconomic instability, high government spending crowding out private investment, significant trade barriers and inadequate infrastructure are all cited as contributing factors. For Pakistan and Nigeria, macroeconomic instability, elite capture and incomplete property rights are frequently mentioned.

Exhibit 2: China Has Converged Fastest Among the World's Most Populous States, But India, Indonesia and Bangladesh Are Also Converging

Real GDP Per Capita in 2022 PPP, log scale (LHS: US = 100, RHS: DM ex-US=100)



Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

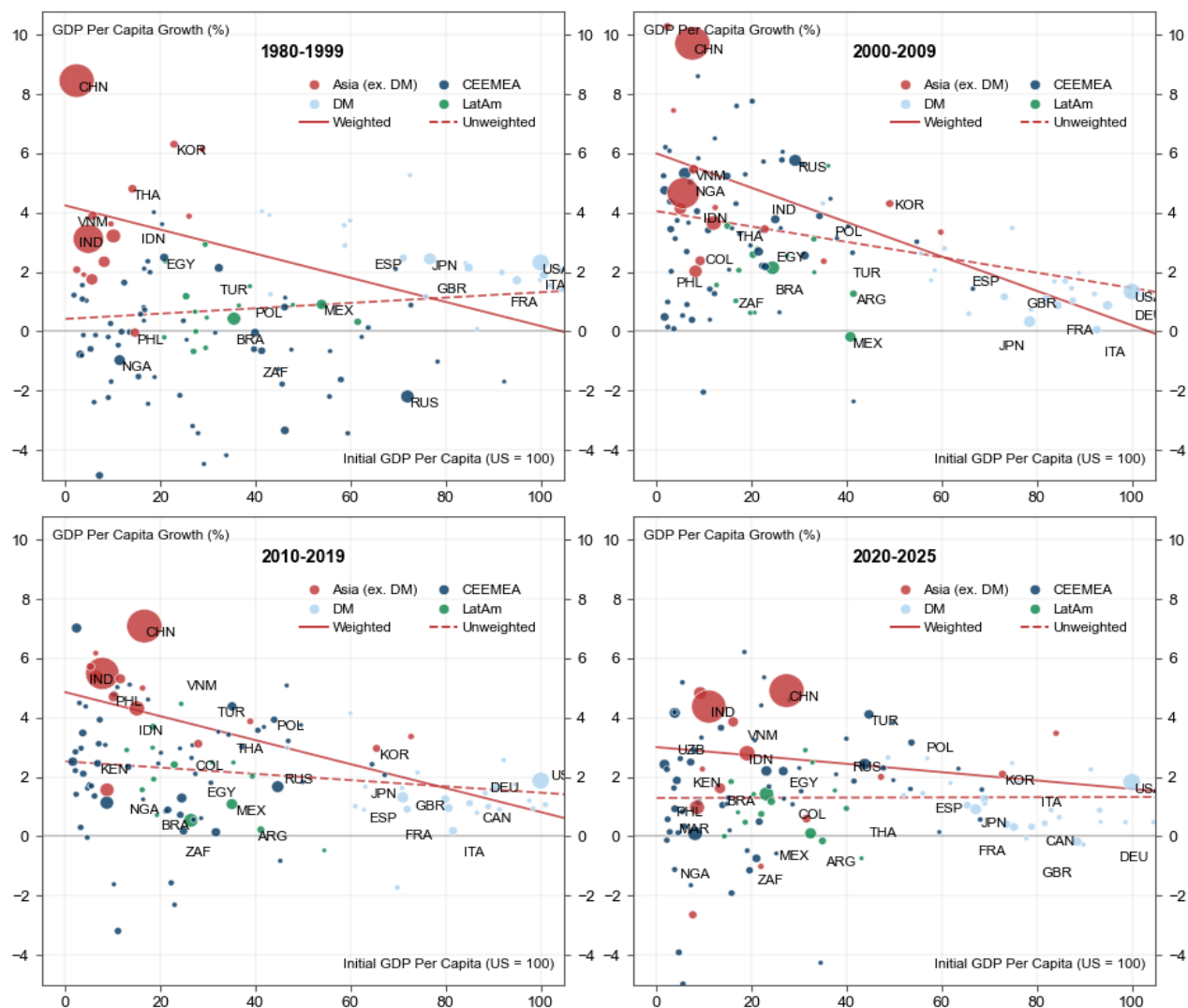
Population-Weighted Convergence is Reasonably Stable Across Decades

Exhibit 3 illustrates average GDP per capita growth in relation to initial GDP per capita levels for EM and DM economies across four distinct periods: 1980-1999, 2000-2009, 2010-2019, and 2020-2025. The level of GDP per capita is shown relative to the US (US = 100). We also show lines of best fit for all 121 economies on both a population-weighted and an unweighted basis.

Across all four periods, the population-weighted regression line slopes downwards, indicating that GDP per capita growth has been higher in low-income economies than in high-income economies (i.e., that there has been convergence). In contrast, the regression lines for unweighted countries are mixed, showing convergence in some periods but not in others. Moreover, the slope of the regression line is steeper for population-weighted countries in all four periods. Note that, although the performance of China and India is central to the narrative of population-weighted convergence – unsurprising given that together they comprise more than a third of the world's population – the regression lines continue to slope downwards from 2000 even when these two countries are excluded.

Exhibit 3: GDP per Capita Has Grown Faster in Low-Income Economies than in High-Income Economies

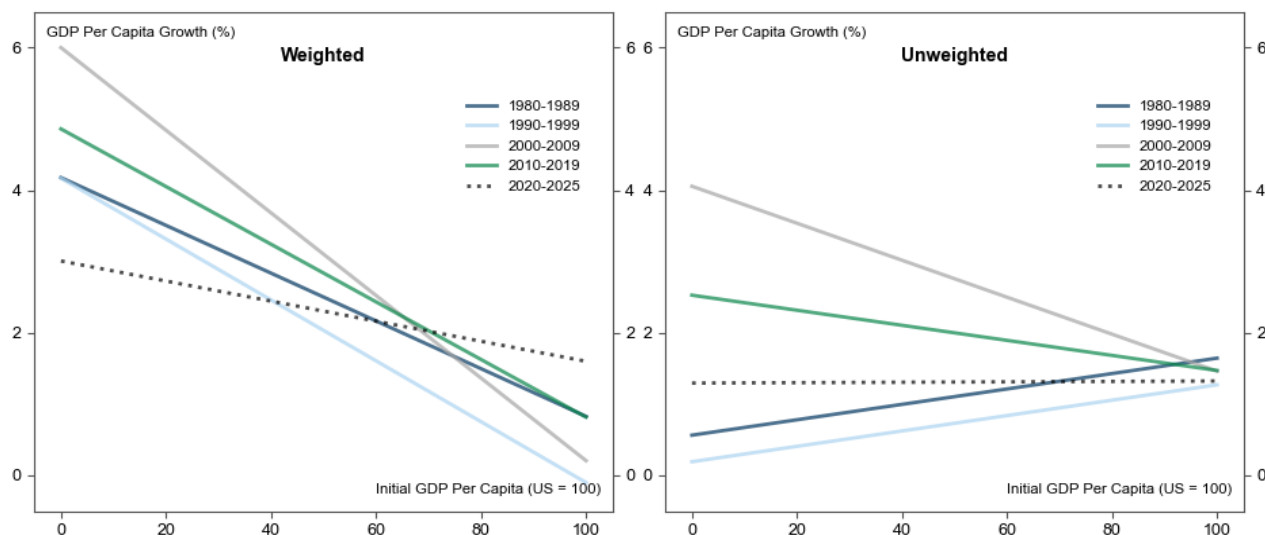
Real GDP Per Capita (PPP, %) vs Initial GDP Per Capita (US=100); Size of bubble = average population size



Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

Exhibit 4 examines decade-by-decade changes in regression slopes. The population-weighted regressions show broadly similar slopes, except for 2020-2025, where the slope is still downward but less marked. The past half decade has seen a series of major shocks – the pandemic, the 2022 energy crisis, and 2025 US tariffs – and it remains to be seen how the slope will look for the decade as a whole. In comparison with the weighted lines, the regression slopes for countries on an unweighted basis display much greater variability.

Exhibit 4: GDP per Capita Growth Has Been Faster in Low-Income Economies in Every Decade on a Weighted Basis, But the Pattern is More Mixed on an Unweighted Basis
Population weighted vs unweighted fitted lines



Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

How rapid is this convergence? Since 2000, individuals in countries with a GDP per capita at 50% of the US level have, on average, experienced GDP per capita growth approximately 2.3pp higher than those in developed economies with productivity levels similar to the US.⁴ Over a 10-year period, GDP per capita levels for such individuals would have risen from 50% of US levels to around 61% of US levels.

How Our Results Compare with the Existing Growth Literature

On one level, our finding that the process of economic convergence is relatively robust and has been a consistent feature since the early 1980s should not be particularly surprising, as it is a central prediction of many economic models.⁵ However, the majority of cross-country studies have found little evidence that poor countries grow faster than rich ones, irrespective of other factors – in other words, there is little evidence of absolute or generalized convergence in the GDP per capita levels of countries over time.⁶

By making the simple adjustment of weighting each country by population size, our results suggest that convergence between poor and rich economies is a more robust feature than implied by previous studies of this type. **If the unit of aggregation is taken to be people, rather than countries – which is what population-weighting achieves – then convergence is alive and well.**

⁴ The slope of the regression line for 2000-2025 is around -0.05.

⁵ Standard (neo-classical) growth theory suggests that relatively poor, low-productivity economies should grow at a faster rate than relatively rich, high-productivity countries. This result follows directly from the assumption of diminishing marginal returns on capital: in developing economies where capital is limited, the returns on capital should be greater than in richer economies where capital is abundant. In turn, the relatively high returns on capital in less developed economies should attract investment, increasing the capital stock and raising living standards.

⁶ A key paper in this literature is Barro and Sala-i-Martin (1992), 'Convergence', *Journal of Political Economy*. Other studies found evidence of 'conditional' convergence – meaning that convergence occurs only when specific factors, such as institutional quality, are present. See Mankiw, Romer and Weil (1992) 'A Contribution to the Empirics of Economic Growth', *Quarterly Journal of Economics*.

We are not the first to adjust for population size in this way. Previous studies have also weighted by population size and, consistent with our findings, have found stronger evidence of convergence on this basis.⁷ Nonetheless, those studies maintain that, when assessing which economic policies drive growth and convergence, it is still reasonable to treat each country as an equal unit, since such policies are largely determined at the national level. We disagree for two reasons: First, economic welfare is a study of people, not nations, and if more people reside in some nations than others, they should be weighted accordingly. Second, a country's size is itself a potentially significant factor influencing growth and convergence. If larger countries tend to grow faster than smaller ones, this may partially be attributable to their size. Overlooking this aspect risks missing a potentially important determinant of growth.⁸

The contrast that we find with unweighted convergence studies suggests an important and intuitive point: periods of rapid globalization, such as 2000–10, tend to be particularly beneficial for the development of small economies with limited domestic markets. However, for large emerging market economies, economic convergence appears to be more resilient to the pace of globalization.

EM Growth Continues to Outstrip DM Growth by a Wide Margin

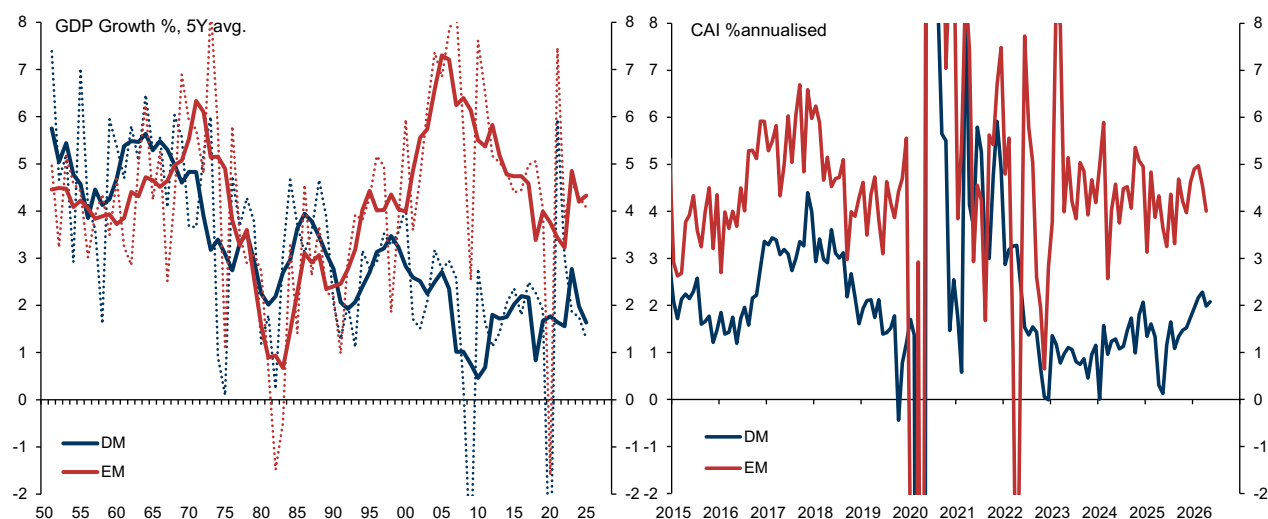
Our findings also explain why EMs consistently outgrow DM countries in our coverage – if convergence were not a fairly consistent feature of the global economy, this would not occur. While population growth is also higher in EMs, it does not fully explain the gap in potential growth rates (4.2% vs 1.6%). Moreover, in the four years since we published our *Path to 2075 long-term economic projections*, the gap between EM and DM growth has so far been larger than we expected (2.6pp vs. 2.0pp).

⁷ In a follow-up paper to Barro and Sala-i-Martin (1992)), Sala-i-Martin (2006) acknowledges that there is stronger evidence of convergence in income distribution at the level of the individual rather than the country but maintains that a country framework makes sense when evaluating which economic policies drive growth and convergence. See Sala-i-Martin (2006), 'The World Distribution of Income: Falling Poverty and... Convergence, Period', *Quarterly Journal of Economics*.

⁸ A large domestic market makes it easier to achieve economies of scale while still focused primarily on domestic demand, especially if industries are protected during their early stages of development. This strategy – protecting fledgling industries until they reach sufficient size – is less likely to succeed in countries with a small domestic market. There is also a related but distinct point concerning economic geography: large economies are able to support bigger cities and benefit from agglomeration effects, and economic convergence is often associated with rising urbanization and increased specialization. This is more difficult to achieve in small economies with small populations.

Exhibit 5: After a Brief Hiatus During the Pandemic, EM Economies Are Outgrowing DM Economies

LHS: GDP Growth (% 5Y avg); RHS: Current Activity Indicator (% change, annualized)



Source: Goldman Sachs Global Investment Research

The ongoing convergence between (population-weighted) poor and rich economies implies that the incomes of EM economies will gradually move closer to those of developed economies – resulting in a further reduction in global income inequality – and that the share of global GDP represented by emerging markets will continue to increase over time.

Looking forward, there are both positive and negative risks to long-term economic growth and convergence. The development of AI seems likely to bolster the strong performance of the US and parts of South-East Asia and lift global GDP productivity growth more generally.⁹ In terms of AI-related capital expenditure, South Korea and Taiwan are already the main beneficiaries, while from a development and adoption standpoint, China seems well-placed to gain.

On the downside, we consider two risks especially significant: First, although global growth and trade have so far been surprisingly resilient to rising tariffs, increased protectionism could still trigger a reversal of globalization. Second, climate change represents a particular threat to Southern Asia and parts of Africa.

Kevin Daly and Alessa Abraham*

**Alessa is an intern in the CEEMEA economics team.*

⁹ For a discussion of the potential boost to EM productivity from AI and technology adoption more generally, see “EM Technology Adoption Cycles and Implications for Artificial Intelligence”, *Global Economics Comment*, 21 May 2026.

Box 1: The Path to 2075 — Long-Term Projections for the Global Economy

The Path to 2075 projections builds on our original BRICs analysis, offering long-term forecasts for the global economy, covering 104 countries out to the year 2075.¹⁰ Exhibit 6 provides a high-level summary of our projections for the major regional aggregates, broken down by decade.

Our projections imply that global growth in 2024-29 (2.8%) will be faster than in 2020-24, but slower than in the 2010-19 decade (3.2%, based on market FX weights).¹¹ We project that EM growth will continue to outstrip DM growth (3.8% vs. 1.8%), with around half of this difference due to demographic factors and the other half due to productivity growth differentials. We expect Asia (ex-DM) to remain the fastest-growing region.

Exhibit 6: A Gradual Slowdown in Global Economic Growth, With EM Growth Continuing to Outstrip DM

	Market FX Weighted								
	2000-2009	2010-2019	2020-2029	2024-2029	2030-2039	2040-2049	2050-2059	2060-2069	2070-2079
World	2.7	3.2	2.4	2.8	2.5	2.1	2.0	1.8	1.7
DM	1.6	1.9	1.5	1.8	1.6	1.4	1.3	1.2	1.1
EM	5.7	5.1	3.6	3.8	3.2	2.6	2.3	2.1	1.9
Asia (ex. DM)	7.6	6.7	4.1	4.2	3.1	2.4	2.1	1.8	1.5
CEEMEA	4.8	3.5	2.6	3.2	3.3	3.1	3.0	2.9	2.7
LatAm	2.8	2.4	2.3	3.0	3.1	2.7	2.3	1.9	1.6

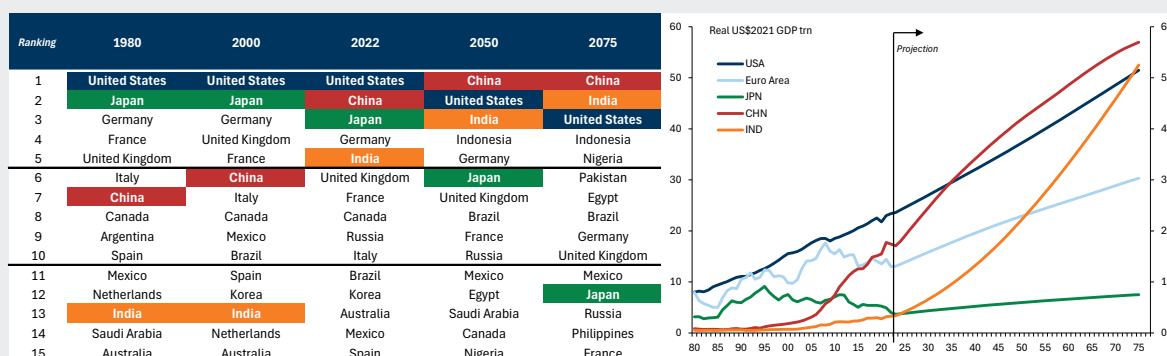
Source: Goldman Sachs Global Investment Research

Exhibit 7 combines our GDP estimates with our long-term real exchange rate projections, to project the real US Dollar value of major economies over time.

In 2020, we project that the world's five largest economies will be China, the US, India, Indonesia, and Germany (with Indonesia displacing Brazil and Russia among the list of largest EMs over this horizon). We estimate that China will overtake the US as the world's largest economy around 2035.¹² If we extend the projection horizon to 2075, the world's three largest economies are China, India, and the US, with India (just) overtaking the US. Interestingly, US potential GDP growth is expected to be materially faster than China's at that horizon because of its better demographic outlook. Given the right policies and institutions, we project that current EMs will make up seven of the world's top ten economies in 2075.

Exhibit 7: Our Projections Imply that China, the US, India, Indonesia, and Germany Will be the World's Five Largest Economies in 2050

World's largest economies (measured in USD)



Source: Goldman Sachs Global Investment Research

¹⁰ "The Path to 2075 — Slower Global Growth, But Convergence Remains Intact", Goldman Sachs Global

Making economic projections over a 50-year horizon for 104 countries inevitably involves a considerable degree of risk and uncertainty. Indeed, given the difficulties in forecasting even one or two years ahead, some readers may be skeptical of the value of forecasts that extend so far into the future. However, one advantage of making long-term projections is that cyclical risk – an important source of short-term forecasting error – tends to mean-revert over time, leaving GDP mostly determined by slower-moving trends in population, capital, and technology.

Economics Paper, 6 December 2022.

¹¹ Because of the large swings in GDP growth generated by the Covid pandemic, the 2024-29 column provides the cleanest indication of what our model indicates for near-term potential growth.

¹² Given the recent pessimism around China's long-term growth prospects, some readers may be surprised that we expect China to overtake the US at this horizon. However, three points are worth bearing in mind in this regard: First, China has already closed most of the gap with US GDP (China's GDP has risen from 12% of the US in 2000 to around 80% currently, measured in market exchange rates). Second, despite significant downward revisions, potential growth in China remains significantly higher than in the US on our revised estimates (4.0% vs. 1.9% for 2024-29). Third, in addition to differences in potential growth, we expect some of the US Dollar's real overvaluation vs. the Chinese Yuan to be unwound over the next 10-15 years.

Box 2: Comparing Our Results Across Different Growth Databases

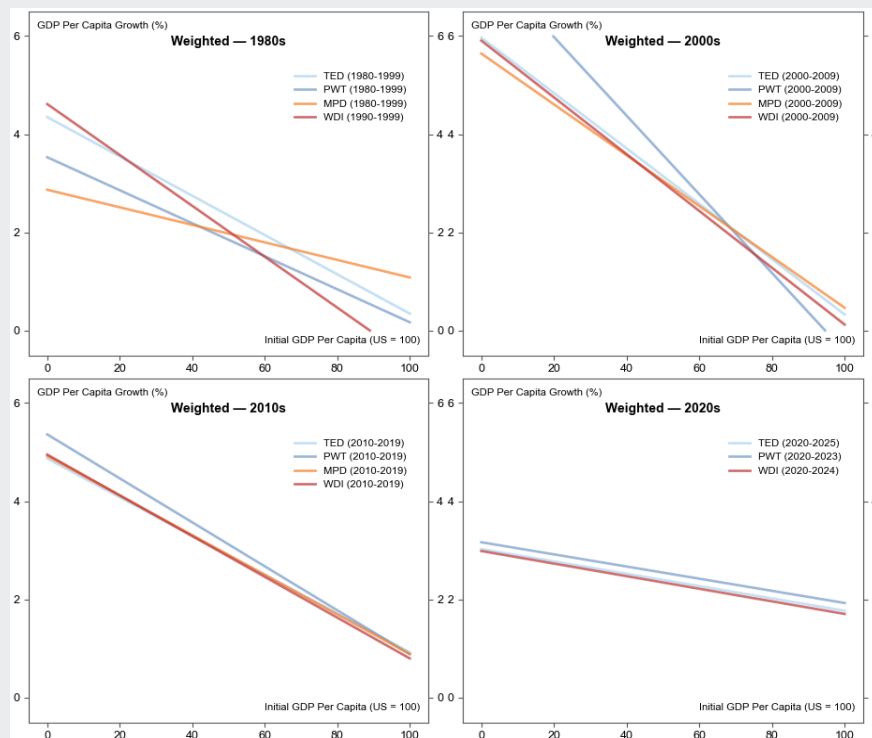
Growth and convergence studies typically use one of four different databases when analyzing long-term economic performance: World Development Indicators (WDI) by the World Bank, the Penn World Tables (PWT), the Maddison Project Database (MPD), and the Total Economy Database (TED) developed by the Conference Board. We generally use the Total Economy Database, partly because it provides the most up-to-date GDP figures. However, to cross-check our findings, we have replicated our analysis using each of these databases to ensure the robustness of our results.

Each of these databases possesses distinct advantages and limitations, making some more suitable than others depending on the research objectives. Among them, the Maddison database offers the most extensive historical coverage, in some cases reaching back centuries, making it ideal for studies in economic history. The World Bank's database has the shortest historical span, beginning only in 1990. Another key distinction is how they adjust GDP figures for differences in Purchasing Power Parity (PPP), which in turn impacts growth indicators.

The results of replicating our convergence analysis across each of the databases are presented in Exhibit 8. Although there are minor variations in the slopes of the fitted lines, the principal result—that weighted convergence persists—remains broadly consistent across all datasets.

Exhibit 8: Our Results Are Similar Across Each of the Main Growth Databases

Fitted lines - weighted by population, across the decades



Source: The Conference Board Total Economy Database™ (April 2026), Maddison Project Database (2020), Penn World Table 11.0, World Bank World Development Indicators, Goldman Sachs Global Investment Research

The Global Economics Team

Jan Hatzius
+1(212)902-0394
jan.hatzius@gs.com
Goldman Sachs & Co. LLC

Joseph Briggs
+1(212)902-2163
joseph.briggs@gs.com
Goldman Sachs & Co. LLC

Sarah Dong
+1(212)357-9741
sarah.dong@gs.com
Goldman Sachs & Co. LLC

Megan Peters
+44(20)7051-2058
megan.l.peters@gs.com
Goldman Sachs International

Disclosure Appendix

Reg AC

I, Kevin Daly, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Kevin Daly Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.